

Harch Intelligence

1,798

GPUS OPERATIONAL

81.5%

RENEWABLE ENERGY

Carbon-aware sovereign GPU cloud delivering 1,798 GPUs across five hubs in Morocco. Industry-leading carbon intensity of ~ 47 gCO₂/kWh — 89% below the global average. Powered by dedicated renewable infrastructure and connected to four submarine cable systems.

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Harch Intelligence

Carbon-Aware AI Infrastructure — Sovereign GPU Cloud

Harch Intelligence operates 1,798 GPUs across five carbon-optimized hubs in Morocco, delivering sovereign AI compute with industry-leading carbon efficiency. Our proprietary carbon-aware scheduling engine routes workloads to the greenest available capacity in real time, achieving a blended carbon intensity of ~47 gCO₂/kWh — 89% below the industry average. Powered by Harch Energy's dedicated renewable infrastructure and connected to four submarine cable systems, Harch Intelligence is the lowest-latency bridge between African compute and European, American, and Middle Eastern markets.

1,798	~47	81.5%	8 ms	500 MW
GPUs Operational	gCO ₂ /kWh Avg	Renewable Energy	Latency to Europe	Pipeline Capacity

Primary Differentiator: Carbon-aware scheduling routes workloads to the greenest hub in real time, achieving ~47 gCO₂/kWh — 89% below the industry average of ~450 gCO₂/kWh. No hyperscaler offers this capability.

GPU Hub Locations

Hub	GPUs	Renewable %	Carbon Intensity	Primary Workload
Harch Ouarzazate	800	97.2%	18 gCO ₂ /kWh	Distributed Training
Harch Dakhla	400	94.8%	32 gCO ₂ /kWh	Sustained Training
Harch Benguerir	350	88.5%	55 gCO ₂ /kWh	Fine-tuning / Batch Inference
Harch Tanger	200	82.1%	95 gCO ₂ /kWh	Low-latency Inference (EU)
Harch Casablanca	48	45%	210 gCO ₂ /kWh	Urban Inference Edge

Table 1: GPU Hub Specifications — All hubs interconnected via 400 Gbps fabric with linear scaling efficiency above 92%.

Technical Specifications

Specification	Value	Notes
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Total Capacity	500 MW Pipeline	Full deployment by 2028
Phase 1	100 MW	Q3 2027
Phase 2	200 MW	Q1 2028
Phase 3	500 MW	Q4 2028
Current GPUs	1,798	5 hubs operational
Target GPUs	100,000+	Full pipeline build-out
PUE	<1.15	Industry-leading efficiency
Latency to Europe	8 ms	Via submarine cable
Cooling	Liquid + Air	Hybrid cooling system
Security	Level IV+	Sovereign-grade
Average Renewable	81.5%	Harch Energy powered
Carbon Intensity	~47 gCO ₂ /kWh	89% below industry average
Backup Power	200 MWh	Battery + Diesel
Network	4 Submarine Cables	Dual diverse landing stations
Energy Cost	\$0.018/kWh at Dakhla	72% below EU average
Interconnect	400 Gbps Fabric	Cross-hub distributed training
SLA	99.99%	Enterprise availability guarantee

Table 2: Technical Specifications — Harch Intelligence Infrastructure

Submarine Connectivity & Network Architecture

Harch Intelligence maintains direct connections to four submarine cable systems, providing ultra-low latency to major global markets. The Dakhla campus sits directly adjacent to the cable landing station, minimizing terrestrial backhaul to under 2 km. Dual diverse landing stations at separate physical locations ensure full path redundancy for all international traffic.

Destination	Latency	Cable System	Redundancy
London / Paris	8 ms	ACE + MainOne	Dual path
Virginia (US East)	35 ms	ACE + Maroc Telecom	Dual path
Dubai (Middle East)	20 ms	MainOne + ACE	Dual path
Lagos (West Africa)	12 ms	ACE + MainOne	Dual path

Table 3: Network Latency — Direct submarine cable connections to global markets

Total cable systems crossing Moroccan waters: 14 systems with capacity exceeding 80 Tbps. This infrastructure positions Harch Intelligence as the lowest-latency bridge between African compute and European, American, and Middle Eastern markets — a connectivity advantage that cannot be replicated by inland European data centers.

Sustainability & Carbon-Aware Scheduling

Carbon-aware scheduling is Harch Intelligence's primary differentiator. Unlike hyperscalers that purchase renewable energy credits or offsets, Harch Energy owns and operates the dedicated solar and wind infrastructure that directly powers each hub. Our proprietary HarchOS scheduler continuously monitors grid carbon intensity across all five sites in real time, automatically routing workloads to the greenest available capacity without requiring any application-level changes.

Metric	Harch Intelligence	Industry Average	Advantage
Carbon Intensity	~47 gCO ₂ /kWh	~450 gCO ₂ /kWh	89% below
Renewable Energy	81.5%	~35%	2.3x higher
Energy Cost (Dakhla)	\$0.018/kWh	\$0.064/kWh (EU avg)	72% cheaper
GPU Training Cost	40-60% below hyperscalers	AWS P5 / GCP A3 / Azure ND	Major savings
PUE	<1.15	1.4-1.6 (industry)	Industry-leading
Free Cooling Hours	~8,500 hrs/year	~2,000 hrs/year (EU)	4x more

Table 4: Sustainability Comparison — Harch Intelligence vs Industry Average

Sovereign Security Framework

Harch Intelligence operates under a sovereign security framework designed for clients who require sovereign-grade data protection. Data residency is guaranteed within Moroccan borders under the Kingdom's data sovereignty regulations, with no data traversing foreign jurisdictions during processing. All access is logged immutably, and compliance artifacts are generated automatically for SOC 2, ISO 27001, and GDPR audit requirements.

Physical Security: Armed perimeter security, biometric access controls at every layer, 24/7 AI-powered surveillance with anomaly detection

Cybersecurity: Real-time DDoS mitigation (multi-terabit capacity), deep packet inspection with intrusion detection/prevention, encrypted data pathways using HSMS

Data Sovereignty: Guaranteed data residency within Moroccan borders, no foreign jurisdiction traversal, immutable access logging

Compliance: Automated artifact generation for SOC 2, ISO 27001, GDPR; zero-trust architecture; regular penetration testing

AI Platform Services

Full-stack AI platform spanning the entire lifecycle from data ingestion and preprocessing through model training, fine-tuning, evaluation, and production deployment. Managed services with enterprise SLAs guaranteeing 99.99% availability and sub-second model serving latency at the edge.

Service	Description	SLA
Training	Distributed training across 800-GPU partitions, PyTorch/TensorFlow/JAX native support	99.99%
Fine-tuning	Managed fine-tuning with hyperparameter optimization, LoRA/QLoRA/PEFT support	99.99%
Inference	Dynamic batching, speculative decoding, automatic model parallelism, cost-per-token optimization	99.99%
MLOps	Pre-built pipelines: experiment tracking, canary deployments, automated rollback, model registry	99.99%
Data Pipeline	Ingestion, preprocessing, versioning, lineage tracking, governance controls	99.99%

Table 5: AI Platform Services — Enterprise SLAs guarantee 99.99% availability across all services

Investment & Financial Overview

Harch Intelligence represents a \$1.14B investment in sovereign AI infrastructure for Morocco and the broader African continent. The project delivers compelling financial returns driven by Morocco's structural energy advantages and the growing global demand for AI compute capacity.

Metric	Value	Detail
Total Investment	\$1.14B	Across 5 hubs + pipeline
5-Year Revenue	\$1.9B	137% TCA
Base IRR	24.7%	Payback in ~30 months
NPV (12%)	\$2,850M	4.6x return

Energy Cost Advantage	40-60% below hyperscalers	\$0.018/kWh at Dakhla
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Table 6: Investment & Financial Overview

Competitive Comparison

Harch Intelligence delivers unique advantages that no hyperscaler or European cloud provider can match. The combination of carbon-aware scheduling, sovereign data residency, and Morocco's structural energy cost advantages creates a value proposition that is structurally impossible to replicate in Europe or North America.

Feature	Harch	AWS	GCP	Azure	OVHcloud
Carbon-Aware Scheduling	Native	No	No	No	No
Sovereign Data Residency	Guaranteed	Regional	Regional	Regional	EU Only
Avg. Carbon Intensity	~47 gCO ₂ /kWh	~450	~400	~420	~350
Renewable Energy	81.5%	~65%	~70%	~70%	~80%
Energy Cost	\$0.018/kWh	\$0.07+	\$0.07+	\$0.07+	\$0.05+
GPU Training Cost	40-60% lower	Baseline	Baseline	Baseline	10-20% lower
Latency to EU	8 ms	1-5 ms	1-5 ms	1-5 ms	1-3 ms
No Egress Fees	Yes	No	No	No	Limited

Table 7: Competitive Comparison — Harch Intelligence vs Major Cloud Providers

Get Started

Harch Intelligence is currently onboarding select enterprise partners for early access to our carbon-aware GPU cloud platform. To request a quote, schedule a technical briefing, or discuss partnership opportunities:

Request a Quote	harch.ai/quote	Custom pricing for your workload
Schedule a Briefing	harch.ai/contact	Technical deep-dive with our team
Join the Team	harch.ai/careers	Build the future of sovereign AI
Developer Docs	harch.ai/docs	API reference, SDKs, and guides